

The Hodge Conjecture via Adaptive Regularization on Currents

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Abstract

The Hodge Conjecture asks whether every rational (k, k) cohomology class on a smooth complex projective variety is algebraic. We reduce it via adaptive regularization on integral currents to a single analytic conjecture: mass-minimizing integral currents in rational (k, k) classes have smooth parts that are complex submanifolds. Mass-minimizers exist by Federer-Fleming compactness; their singular sets have Hausdorff codimension ≥ 2 by Almgren-De Lellis-Spadaro; once the smooth part is complex, algebraicity follows from Chow's theorem. The calibration lower bound shows that a mass-minimizer is complex if and only if it achieves the Kähler calibration bound, but forcing this achievement from the cohomology class alone is the open step.

1 Setup

Let X be a smooth complex projective variety of complex dimension n , with Kähler form ω . A *Hodge class* is $\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$.

Hodge Conjecture. Every Hodge class is algebraic: $\gamma = \sum_i q_i [Z_i]$ for irreducible complex subvarieties $Z_i \subset X$ of codimension k and $q_i \in \mathbb{Q}$.

2 Calibration

Lemma 1 (Calibration lower bound). *For any integral $(2k)$ -current T on X representing a class $[T]$:*

$$M(T) \geq T\left(\frac{\omega^k}{k!}\right) = \left\langle [T], \left[\frac{\omega^k}{k!}\right] \right\rangle.$$

Proof. The comass $\|\omega^k/k!\|_\infty = 1$ on a Kähler manifold [1]. Hence $T(\omega^k/k!) \leq M(T) \cdot \|\omega^k/k!\|_\infty = M(T)$. $\square$

Proposition 2 (Harvey–Lawson equality condition [1]). *An integral current T satisfies $M(T) = T(\omega^k/k!)$ if and only if T is calibrated by $\omega^k/k!$, which holds if and only if the smooth part of T is a (positively oriented) complex k -submanifold.*

Corollary 3. *If a mass-minimizer T_0 in a Hodge class γ achieves $M(T_0) = \langle \gamma, [\omega^k/k!] \rangle$, then T_0 is calibrated and its smooth part is a complex submanifold.*

Remark 4 (Route A is circular). Corollary 3 gives: *if the mass-infimum in class γ equals the calibration lower bound, then T_0 is complex.* But the mass-infimum equals the lower bound if and only if the class contains a calibrated representative, i.e. a complex subvariety with class γ . That is the Hodge Conjecture itself. Route A via calibration does not provide an independent proof.

3 Adaptive Regularization and Compactness

Definition 5. For $\alpha > 0$ and Hodge class γ , define on integral $(2k)$ -currents representing γ :

$$E_\alpha[T] = M(T) + \alpha \cdot M(T)^2.$$

Theorem 6 (Federer–Fleming compactness [2]). *The space of integral currents with bounded mass and zero boundary is compact in the flat norm topology.*

Lemma 7 (Mass-minimizers exist). *For each $\alpha \geq 0$, there exists a mass-minimizing integral current T_α representing γ , with $M(T_\alpha) \leq M(T_{\text{ref}})$ for any fixed reference current T_{ref} in class γ .*

Proof. Minimizing sequences are precompact by Theorem 6; E_α is lower semicontinuous in the flat norm; existence follows by the direct method. $\square$

Lemma 8 (Uniform mass bound). *$M(T_\alpha) \leq M(T_{\text{ref}})$ uniformly in α , where T_{ref} is any fixed current representing γ .*

Proof. Since T_α minimizes E_α : $M(T_\alpha) + \alpha M(T_\alpha)^2 \leq M(T_{\text{ref}}) + \alpha M(T_{\text{ref}})^2$. The bound on $M(T_\alpha)$ follows (it cannot exceed $M(T_{\text{ref}})$ for any $\alpha \geq 0$). $\square$

Theorem 9 (Convergence to mass-minimizer). *As $\alpha \rightarrow 0^+$, $T_\alpha \rightarrow T_0$ in the flat norm along a subsequence, where T_0 is mass-minimizing in class γ .*

Proof. The uniform mass bound and Theorem 6 give precompactness. Any limit is mass-minimizing by lower semicontinuity of mass. $\square$

4 Regularity of Mass-Minimizers

Theorem 10 (Almgren regularity [3], De Lellis–Spadaro [4]). *A mass-minimizing integral current T_0 on a Riemannian manifold is smooth away from a closed singular set $\Sigma \subset \text{spt}(T_0)$ with $\mathcal{H}^{2k-2}(\Sigma) = 0$ (Hausdorff codimension ≥ 2).*

Lemma 11 (Smooth implies algebraic, given complex). *Suppose the smooth part of T_0 (away from Σ) is a complex k -submanifold of the projective variety X . Then $T_0 = [Z]$ for an algebraic subvariety $Z \subset X$.*

Proof. By Theorem 10, Σ has codimension ≥ 2 , so the smooth part of T_0 is a complex manifold away from a thin set. Its closure is a compact complex analytic subvariety of X . By Chow’s theorem [5], every compact complex analytic subvariety of a projective variety is algebraic. $\square$

5 The Open Step

The entire argument reduces to one conjecture:

Conjecture 12 (Complex regularity conjecture, CRC). *Let X be a smooth complex projective Kähler manifold and $\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$ a Hodge class. Let T_0 be a mass-minimizing integral current representing γ . Then the smooth part of T_0 is a complex k -submanifold.*

Theorem 13 (Hodge Conjecture, conditional on CRC). *Assume Conjecture 12. Then every Hodge class is algebraic.*

Proof. By Lemma 7, a mass-minimizer T_0 exists. By CRC, its smooth part is complex. By Lemma 11, $T_0 = [Z]$ for an algebraic cycle Z . Therefore $\gamma = [T_0] = [Z]$ is algebraic. $\square$

6 Evidence for CRC and the Gap

What is known.

- If the mass-minimizer achieves the calibration bound $M(T_0) = \langle \gamma, [\omega^k/k!] \rangle$, then CRC holds by Harvey-Lawson (Proposition 2). This is the calibration route, identified as circular in Remark 4.
- For $k = 1$ (divisors), Hodge is known (Lefschetz $(1, 1)$ theorem); CRC holds because mass-minimizing 1-currents in $H^{1,1}$ are calibrated.
- For $k = n - 1$ (top codimension), Hodge is known by Poincaré duality.
- The singular set Σ has measure zero, so it does not affect the cohomology class. The argument runs on the smooth part alone.

The gap. CRC requires showing that the cohomology constraint $[T_0] \in H^{k,k}$ forces the tangent planes at smooth points to be complex k -planes. This is a condition on the current geometry, not a consequence of the $\partial\bar{\partial}$ -lemma (which applies to forms, not currents). Making this precise — extending the Harvey-Lawson theory to singular mass-minimizers in (k, k) classes without assuming calibration a priori — is the remaining work.

Remark 14 (Template summary). • *Regularize:* $E_\alpha[T] = M(T) + \alpha M(T)^2$; minimizers exist for all α . ✓

- *Bound:* $M(T_\alpha) \leq C$ uniformly. ✓
- *Compact:* Federer-Fleming. ✓
- *Regularity:* smooth part of T_0 is complex. Conditional on CRC.

References

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